

CYBERSECURITY ACTIVITY · COLLEGE EDITION

SOCIAL ENGINEERING ATTACKS

How human trust becomes the first step in a cyber breach.

Original 12-page infographic PDF for a college cybersecurity activity.

01 · THE OPENING MOVE

The attack starts with a believable story

Social engineering manipulates people into revealing information, clicking, approving, transferring, or opening access.

Attackers rarely begin with a dramatic technical exploit. They begin with a situation that feels ordinary enough to trust.

> Urgency, authority, fear, curiosity, and helpfulness become leverage.

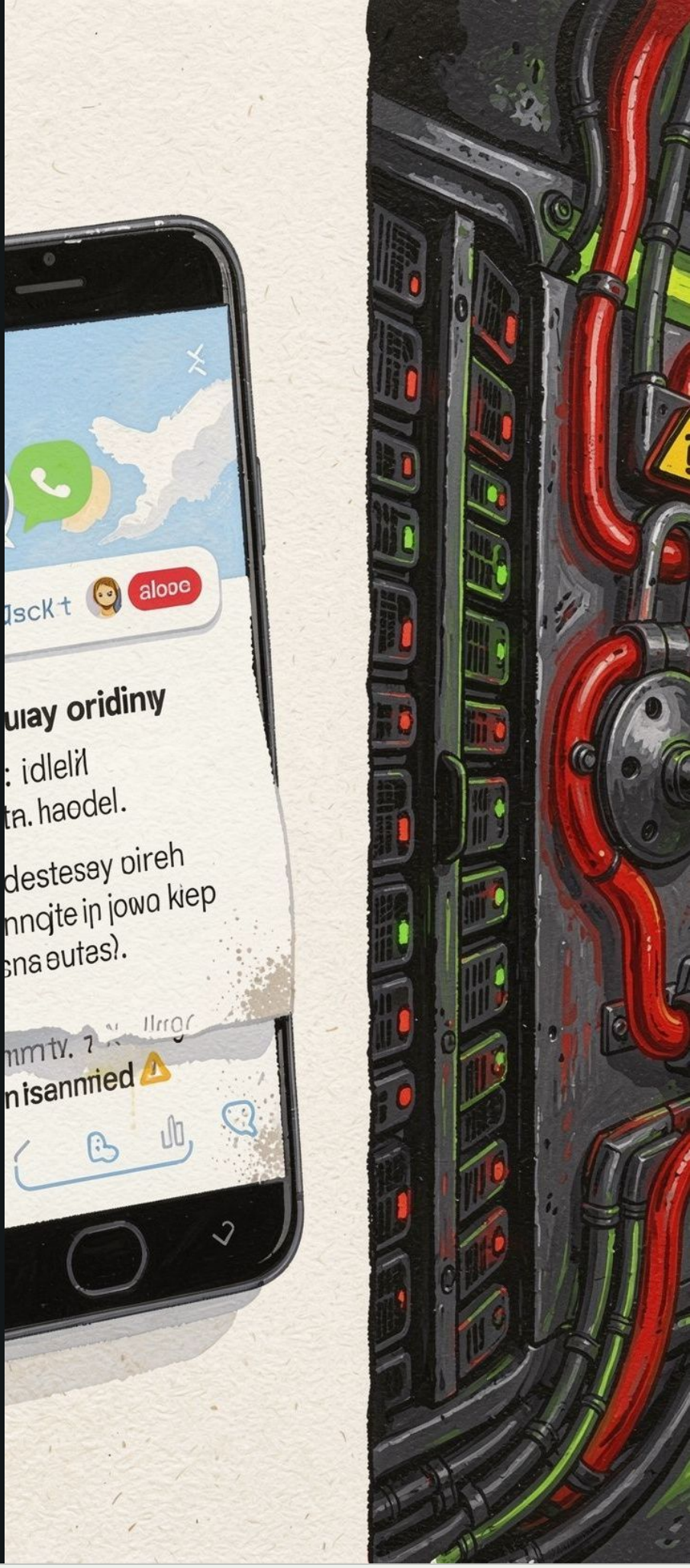


Fig. 1 A familiar story can conceal an unfamiliar destination.

The manipulation loop

Each step makes the next one feel easier to accept.

1. **Hook:** a message, call, reward, or warning grabs attention.
2. **Trust:** the attacker imitates a person, brand, colleague, or support team.
3. **Action:** the target clicks, shares, approves, downloads, or pays.
4. **Access:** credentials, device access, or internal permissions are gained.
5. **Impact:** data theft, account takeover, malware, fraud, or disruption follows.

⚠ **Decision point:** the safest interruption is often before the first action, while the story is still being tested.

Phishing: the fake message trap

A copied logo can look familiar while the destination is hostile.

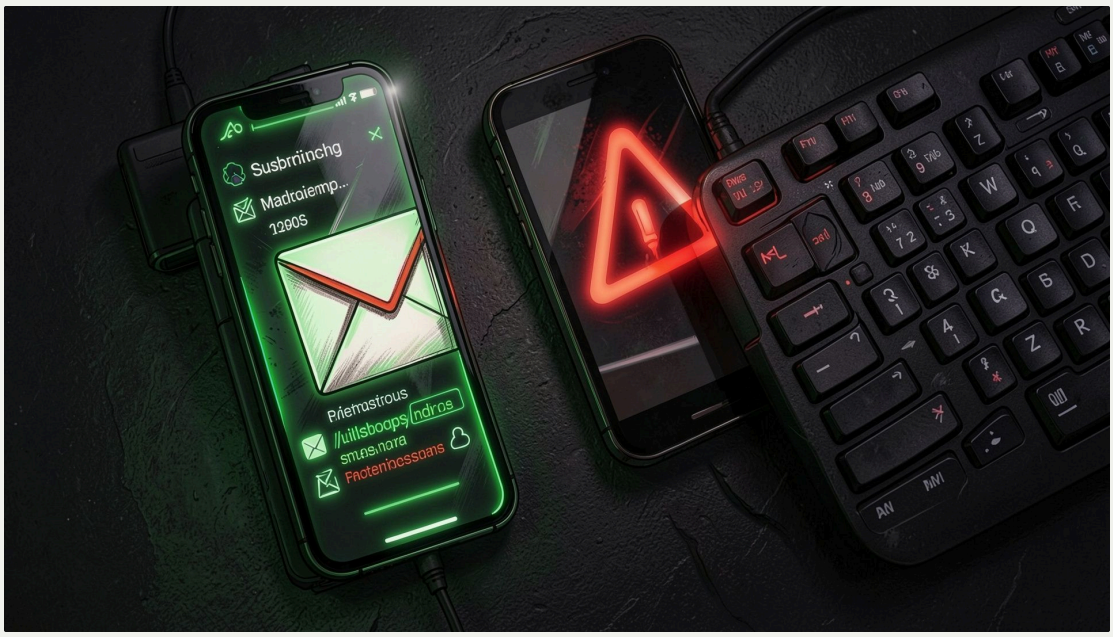


Fig. 2 The message asks for confidence before it earns verification.

Red flags

- Urgent language or unusual sender details
- Mismatched links or unexpected attachments
- Requests for passwords, payment, or secrecy

Safer response

Hover before clicking. Open the official site separately. Use MFA. Report suspicious messages instead of forwarding them.

Pretexting and baiting: trust with a costume

THE BELIEVABLE STORY

Pretexting

The attacker invents a role or situation: IT support, a manager, a delivery agent, or a bank representative.

> Verify through a known channel before sharing information or changing access.

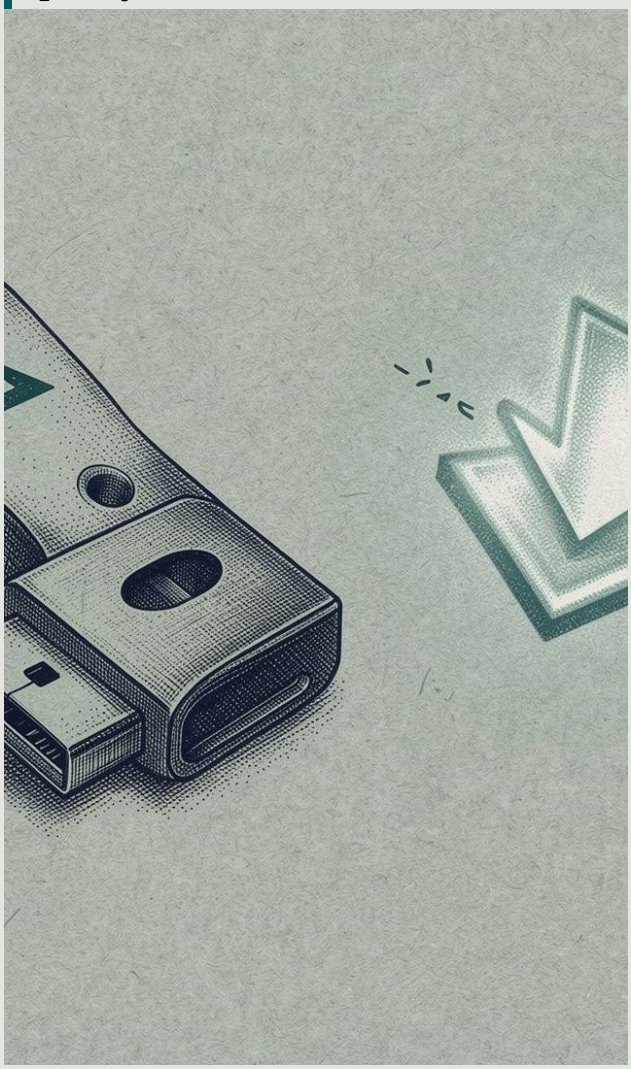


THE TEMPTING REWARD

Baiting

The attacker offers something desirable: a free download, USB drive, giveaway, or exclusive file.

> Curiosity is not evidence. Never let a tempting object bypass security policy.



Target: the third-party foothold

A trusted partner can become the first door into a much larger system.

VENDOR

HVAC access

Attackers used access connected to a third-party HVAC vendor to enter Target's network.

INTERNAL PATH

Payment systems

The compromise moved through a chain of access and payment systems.

OUTCOME

Data extraction

The chain led to data extraction.

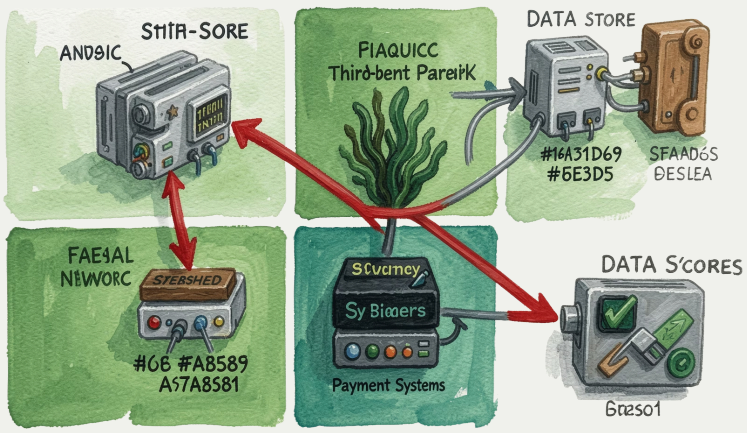


Fig. 3 Prevention lens: vendor risk reviews, least privilege, network segmentation, monitoring, and faster containment.

One attachment, many weak points

A single employee action became more damaging because other protections did not hold.

1
Email
An employee was tricked into opening a malicious attachment.

2
Credentials
Administrator credentials were exposed.

3
Files
Protection around sensitive files was weak.

4
Disruption
Destructive malware took systems offline and disrupted operations.



Fig. 4 Prevention lens: phishing resistance, MFA, password hygiene, encryption, segmentation, and tested recovery.

07 · CASE STUDY / TWITTER

Twitter: when the help desk becomes the target

Attackers used phone calls while pretending to be Twitter's IT department. They gained internal access and took over high-profile accounts.

> Technical controls can be bypassed when support processes trust the wrong voice.



Verify identity

Use a known record, not the caller's details. Confirm through an out-of-band channel.

Restrict access

Restrict admin tools; require approval for resets and sensitive changes.

Log actions

Keep detailed logs to make unusual access visible.

CREATED BY: KENIL PATEL

What the cases have in common

The breach was social at the entry point, but layered failures made the impact larger.

ENTRY POINT

Trust was weaponised

Vendor access, an employee action, or a support interaction opened the path.

MISSED CONTROL

One weakness was not alone

Access control, monitoring, segmentation, or recovery also failed to limit damage.

BETTER CONTROL

Prevention is layered

People, process, and technology must reinforce each other.

Fig. 6 Strong security is a chain of reinforcing decisions, not one perfect barrier.

Break the chain before the breach

1. **Pause:** urgency is a signal, not a reason to skip checks.
2. **Verify:** use a known phone number, official portal, or second person.
3. **Protect:** use MFA, unique passwords, least privilege, segmentation, and encryption.
4. **Report:** preserve the message and escalate quickly.
5. **Learn:** update the process so the same story fails next time.



Fig. 7 The safest story is the one you verify before you act.

THANK YOU